

Mini Lesson Sample Script

Materials: Personal white board

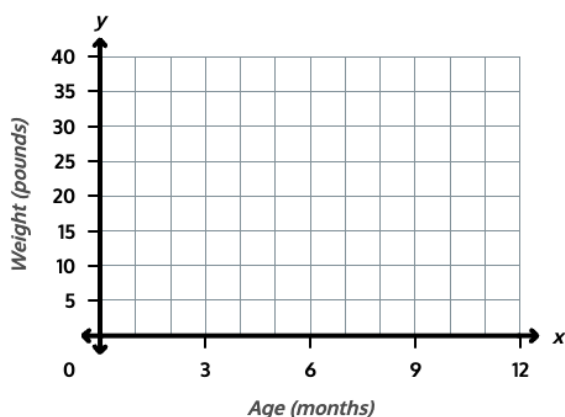
1. Plot data from a table onto a line graph and connect the points.

This table shows how much a puppy named Rosie weighed at different ages.

Age (months) Weight (pounds)

0	4
3	15
6	25
9	32
12	35

Plot each point from the table on the coordinate plane, then connect the points with line segments to form a line graph.



Answer:

T: Take a look at this coordinate plane. What does the x-axis show?

S: The puppy's age in months.

T: And the y-axis?

S: Her weight in pounds.

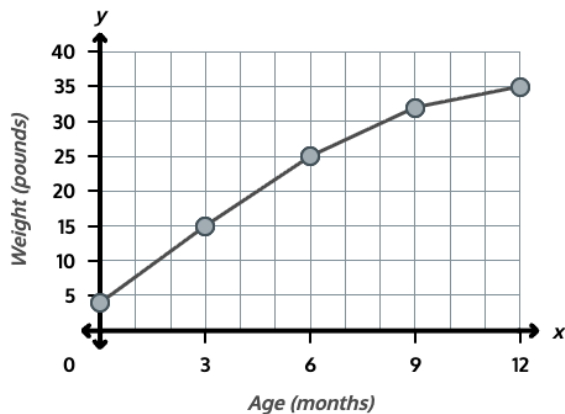
T: Look at the gridlines on the y-axis. Each gridline goes up by how much?

S: 5 pounds.

T: Right. And on the x-axis?

S: 3 months between each line.

T: Let's plot the first point together. At 0 months, Rosie weighed 4 pounds. I'll put a point at (0, 4), right here. Now your turn for 3 months. Rosie weighed 15 pounds. Where does that point go?



S: Over 3 months and up to 15 pounds. (Plots point.)

T: Plot the rest of the points from the table, then connect each pair of points with a line segment.

S: (Plots and connects points.)

2. Analyze the line graph to find changes between intervals and identify when values stayed constant.

Use Rosie's line graph from Problem 1 to answer the questions.

a. How much weight did Rosie gain between 0 months and 3 months?

T: How can you figure out how much weight Rosie gained between 0 months and 3 months?

S: I subtract. 15 pounds minus 4 pounds is 11 pounds.

b. How much weight did Rosie gain between 9 months and 12 months?

T: Check that on the graph by counting up the y-axis from 4 to 15.

c. During which 3-month interval did Rosie gain the most weight?

S: 5, 10, 11. Yep, 11 pounds.

d. Suppose Rosie weighed 35 pounds at 15 months and 35 pounds at 18 months. What would the line look like between 12 months and 18 months, and what does that tell you about Rosie's weight?

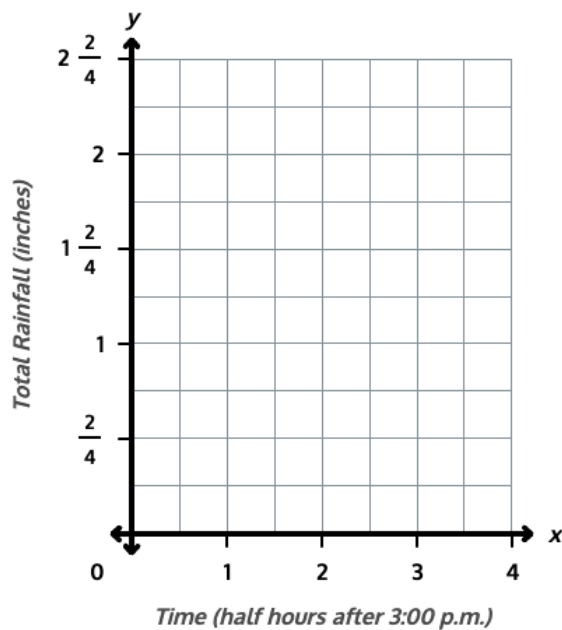
T: Now find the weight gained between 9 months and 12 months.

S: 35 minus 32 is 3 pounds.

<i>Answer: a) 11 pounds; b) 3 pounds; c) 0 to 3 months; d) The line would be horizontal (flat), showing that Rosie's weight stayed the same.</i>	T: Look at each segment on the graph. Which 3-month interval was Rosie's biggest growth spurt? S: 0 to 3 months. That line is the steepest. T: Right. The steeper the segment, the more weight she gained in those 3 months. Now think about this. What if Rosie weighed 35 pounds at 15 months and still 35 pounds at 18 months? What would the line look like between 12 months and 18 months? S: It would be flat. T: What does a flat line mean about her weight? S: She didn't gain any weight. Her weight stayed the same.
3. Plot a new data set on a line graph and identify the interval with the greatest change by comparing the steepness of line segments.	
This table shows the total rainfall during a storm, measured each half hour.	T: Plot the five points from the table, connect them with line segments, and find the half hour with the most rainfall. S: (Works.) T: Which half hour had the most rain, and how do you know?

Time	Total Rainfall (inches)
3:00 p.m.	0
3:30 p.m.	$\frac{1}{2}$
4:00 p.m.	$1\frac{1}{4}$
4:30 p.m.	$1\frac{1}{4}$
5:00 p.m.	2

Plot each point on the coordinate plane and connect them with line segments. Then identify the half-hour interval during which the most rain fell.



Answer: 3:30 p.m. to 4:00 p.m. ($\frac{3}{4}$ inch of rain fell during this half hour, more than any other interval).

S: 3:30 to 4:00. That segment is the steepest.

T: Confirm with subtraction.

S: $1\frac{1}{4}$ minus $\frac{1}{2}$ is $\frac{3}{4}$ inch. That's more than any other half hour.